Heriberto Patino Luna

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EDUCATION

University of Wisconsin–Whitewater, Whitewater, WI | Expected Graduation: May 2027 | GPA: 3.8

B.S. Physics and B.S. Geography (Geology Emphasis) | Mathematics & Chemistry Minors

RESEARCH EXPERIENCE

Degradation of PFAS Through Photocatalysis with Semiconductor Materials

Advisors: Dr. Paul House, Dr. Bianchi Mendez, & Dr. Pedro Hidalgo

Universidad Complutense de Madrid & University of Wisconsin-Whitewater | May 2026 - Present

- Synthesized MoO_3 micro and nanostructures through the Joule heating method while applying an electric field perpendicular to molybdenum wire axis.
- Maintained the experimental apparatus, including soldering the copper electrodes for the applied electric field.
- Characterized the synthesized MoO_3 micro and nano structures using scanning electron microscopy (SEM), micro-Raman spectroscopy, and XRD.
- Developed Python scripts to generate plots of micro-Raman spectroscopy and XRD data.
- Analyzed COMSOL Multiphysics simulations of the Joule heating synthesis process, which consisted of the coupling of electrostatics, electric currents, heat transfer, and structural mechanics.

Calculating Mutual Information from Mitral Cell Activity and Stimulus for the Investigation of Granule Cells on the Olfactory Bulb.

Advisor: Dr. Martin Tchernookov

University of Wisconsin-Whitewater | August 2024 - Present

- Performed numerical simulations of a system of Langevin equations in Python to model mitral cell activity in the olfactory bulb.
- Developed Python scripts to compute autocorrelation of mitral cell activity as a precursor to mutual information calculation.

Simulation and Control of Spin Waves in Chromium Sulfur Bromide Using Finite Difference Time-Domain Method and Laser-Cut Interfaces

Advisors: Dr. Jin Chenhao & Samuel Brantly

University of California Santa Barbara | Jun 2025 – August 2025

- Collaborated with two undergraduate researchers in the development of two-dimensional finite-difference time-domain (FDTD) and finite-difference frequency-domain (FDDF) simulations based on the dispersion relation of spin waves in CrSBr reported in the literature, allowing the comparison with pump-probe experimental measurements.
- Assembled a laser-microscope system for the fabrication of custom interface geometries in CrSBr samples.
- Developed Python scripts to automate the interface fabrication by controlling the stage where the sample was placed with Thorlabs motors. Geometry files drawn in KLayout were used as motion paths that enabled the fabrication of the desired interface geometry.

Basaltic Lava Flow Rheology of Core Samples from Pacaya Volcano, Guatemala to Parameterize Volcanic Flank Stability Models.

Advisors: Dr. Tushar Mittal, Dr. Christelle Wauthier, & Raphael Affinito

The Pennsylvania State University | May 2024 – Present

- Shadowed cyclic uniaxial deformation experiments performed to characterize the mechanical behavior of the basaltic lava flow.
- Processed experimental data using MATLAB and Python, including calculating P-wave velocities from lead zirconate titanate (PZT) transducer measurements, axial stress from load-cell data, and strain from DC displacement transducer (DCDT) measurements.
- Developed Python scripts to generate stress-strain plots and qualitatively characterize mechanical behavior of the material, including plastic deformation and creep.
- Assisted in the implementation and evaluation of linear and nonlinear elastic models, as well as fixed and changing viscosity viscoelastic models using Python

- Contributed to the development of a parameter optimization framework in Python that utilized Optuna for acquiring initial guesses that were used to initialize the Markov Chain Monte Carlo (MCMC) sampler to estimate the full posterior distribution.
- The elastic moduli and viscosity were defined as non-linear functions of experimental data, and parameters that were calibrated with the parameter optimization framework.
- Research completed; manuscript currently in preparation for publication.

Slope Failure in the Lab with Sand and Water Using a Multimodal Monitoring Device for Data Collection.

Advisors: Dr. Prajukti (juk) Bhattacharyya & Dr. Ozgur Yavuzcetin

University of Wisconsin-Whitewater | Winter 2023 – Fall 2024

- Designed and conducted small-scale physical modeling experiments to optimize the detection of creep behavior and micro and macro-failures using a multimodal monitoring system consisting of a load cell, moisture, temperature sensors.
- Analyzed experimental data using IGOR Pro to identify creep behavior and micro and macro-failure signatures during the experiments.
- Developed Python scripts to automate data processing and accelerate analysis procedures.

TEACHING EXPERIENCE

Tutor - University of Wisconsin-Whitewater (Physics Department) | January 2025 – present

- Provided one-on-one and small-group instruction for introductory Physics I and II courses.

Tutor - University of Wisconsin-Whitewater (Math Success Center) | February 2025 – December 2025

- Provided one-on-one and small-group tutoring in College Algebra, Trigonometry, Calculus I–III, Linear Algebra, and Quantitative Analysis.

Teaching Assistant – University of Wisconsin-Whitewater | January 2024 – December 2025

- Assisted students in the learning process and setting up/cleaning labs for Introduction to Geology & Physical Geography.

WORK EXPERIENCE

Environmental Technician - Beloit, WI | May 2023 - August 2023

Brownfield Environmental & Engineering

- Conducted on-site monitoring during earthwork using a photoionization detector (PID) to assess volatile organic compound (VOC) emissions from excavated soil.
- Prepared daily field reports documenting earthwork operations, PID measurements, and monitoring locations using site plans and field observations.
- Researched historical land-use information of assigned properties with the goal to determine if there was an environmental threat.

SELECTED COURSEWORK

Study between the relationship of water temperature and Geyser Eruption Periodicity (Physics Travel Study – Iceland)

Advisor: Dr. Ozgur Yavuzcetin

University of Wisconsin-Whitewater | Spring 2024

- Utilized a small-scale physical model of a geyser system consisting of a heated reservoir, conduit, and surface pool analog.
- Developed a Raspberry Pi-based temperature monitoring system and Python scripts to process temperature data and analyze eruption periodicity of the small-scale physical modeling experiments.
- Collected field measurements in Iceland using thermal imaging and timing observations of geyser eruptions.
- Identified a correlation between water temperature and eruption periodicity, while recognizing the influence of additional parameters such as conduit geometry as noted by the literature.

TECHNICAL EXPERIENCE

Laboratory Techniques

- Scanning electron microscopy (SEM), micro-Raman spectroscopy, infrared (IR) spectroscopy, fluorescence spectroscopy, UV-Vis spectroscopy, and Portable X-ray florescence (pXRF).

Fabrication

- Soldering, Raspberry Pi systems, MIG welding, and stick welding.

Programming & Software:

- Python, COMSOL Multiphysics, MATLAB, IGOR Pro, KLayout, and Mathematica.

SCHOLARSHIPS & AWARDS

- URiCA Summer Fellowship *(2026).*
- Barry Goldwater Scholarship *(2025).*
- McNair Scholars Program *(2023–Present).*
- American Institute of Professional Geologists (AIPG) Scholarship *(2024).*
- University of Wisconsin–Whitewater Departmental Scholarship *(2023).*
- King/Chávez Scholarship *(2022–2023).*

VOLUNTEERING

TRIO and Upward Bound Program – University of Wisconsin-Whitewater | February 2026 & February 2025

- Presented current research project, spin-wave propagation in CrSBr in 2026 and basaltic lava flow rheology in 2025, to high school students and led small-group discussions about the research process and college opportunities.

Physics Outreach - University of Wisconsin-Whitewater | June 2025

- Delivered a presentation to high school students on the physics and technological applications of the research project on spin-wave propagation in CrSBr, with the goal of motivating interest in college and STEM careers.
- Developed hands-on activities using iron filings and magnets introduce the concept of magnetic fields.

Science Fair – Fort Atkinson Elementary School | April 2026 & April 2025

- Participated as a judge of research posters from elementary school students.
- Gave encouragement and written feedback based on the rubric provided by organizers.

Boxes & Walls Event - University of Wisconsin-Whitewater | October 2024

- Volunteered in the Latinos room, engaging students and community members in discussions about dismantling stereotypes and addressing oppression.

Club Involvement - University of Wisconsin-Whitewater | October 2023

- Delivered an oral presentation emphasizing the positive impact of undergraduate research to encourage members of the Environmental Science club to get involved.

Stream Monitoring - University of Wisconsin-Whitewater | September 2023

- Went to a local stream and collected water velocity, depth, and pH data for the school's sustainability office.

Grandchildren's University: Geology Outreach – University of Wisconsin-Whitewater | July 28, 2023

- Assisted kids and their grandparents in the learning process about the rocks and minerals displayed.

PRESENTATIONS

Professional Societies Conference Presentations

- **Patino Luna, H**., Affinito, R., Wauthier, C., Marone, C., & Mittal, T. Rheology of Basaltic Lava Flows from Pacaya with Stress-Strain Steps: A Laboratory Approach. Poster session presented at: American Geophysical Union; 2024, December 12th, Thursday; Washington D.C.
- **Patino Luna H**., & Larson C., Utilizing a Load Cell Sensor to Collect Data on Slope Failure from Small-scale Physical Modeling Experiments. Poster session presented at: Geological Society of America's Connects; 2024, September 23, Monday; Anaheim, CA.
- **Patino Luna H**., Stephens B., Larson C., Lambert J., Wink D., & McIntosh M. Simulating Conditions of Slope Failure in the Lab with Sand and Water Using a Multimodal Monitoring Device for Data Collection. Poster session presented at: Geological Society of America's Connects; 2023, October 16th, Monday; Pittsburg, PA.

Academic Conference Presentations

- **Patino Luna H**., Brantly S., Cao. J, Yin K., & Chenhao J. Simulation and Control of Spin Waves in CrSBr Using FDTD and Laser-Cut Interfaces. Oral presentation at: University of California Santa Barbara; 2025, August 18th, Monday; Goleta, CA.
- **Patino Luna H**., Brantly S., Cao. J, Yin K., & Chenhao J. Simulation and Control of Spin Waves in CrSBr Using FDTD and Laser-Cut Interfaces. Poster presentation at: UCSB Summer Undergraduate Research Symposium; 2025, August 14th, Thursday; Goleta, CA.

- **Patino Luna H**., Affinito R., Wauthier C., Marone C., & Mittal T. Investigating Basaltic Lava Flow Rheology of Core Samples from Pacaya Volcano, Guatemala to Parameterize Volcanic Flank Stability Models. Poster session presented at: Baylor University McNair Scholars Research Conference; 2024, August 2nd, Friday; Waco, TX.
- **Patino Luna H**., Affinito R., Wauthier C., Marone C., & Mittal T. Investigating Basaltic Lava Flow Rheology of Core Samples from Pacaya Volcano, Guatemala to Parameterize Volcanic Flank Stability Models. Oral presentation at: Office of Graduate Educational Equity Programs; 2024, July 27th**,** Monday; State College, PA.
- **Patino Luna H**., Stephens B., Larson C., Wink D., & McIntosh M. Simulating Conditions of Slope Failure in the Lab with Sand and Water Using a Multimodal Monitoring Device for Data Collection. Poster session presented at: National Conference of Undergraduate Research; 2024, April 9th, Tuesday; Long Beach, CA.
- **Patino Luna H**. Analyzing Shoreline Erosion and Accretion of the Lake Michigan Shore Using LIDAR. Oral presentation at: Upper Midwest Honors Conference; 2024, April 7th, Sunday; Dubuque, IA.
- **Patino Luna H**., Stephens B., Larson C., Wink D., & McIntosh M. Simulating Conditions of Slope Failure in the Lab with Sand and Water Using a Multimodal Monitoring Device for Data Collection. Poster session presented at: Spring Research Day Symposium; 2024, March 21st, Thursday; Whitewater, WI.
- **Patino Luna H**., Stephens B., Larson C., Wink D., & McIntosh M. Simulating Conditions of Slope Failure in the Lab with Sand and Water Using a Multimodal Monitoring Device for Data Collection. Poster session presented at: Research at the Rotunda; 2024, March 6th, Wednesday; Madison, WI.
- Larson C., & **Patino Luna H**. Landslide Detection System Using RPi and Strain Gauge. Poster session presented at: Fall 2023 Research Day Symposium; 2023, November 9th, Thursday; Whitewater, WI.
- **Patino Luna H**., Stephens B., Larson C., Lambert J., Wink D., & McIntosh M. Simulating Conditions of Slope Failure in the Lab with Sand and Water Using a Multimodal Monitoring Device for Data Collection. Poster session presented at: Fall 2023 Research Day Symposium; 2023, November 9th, Thursday; Whitewater, WI.
- **Patino Luna H.**, Stephens B., Larson C., Lambert J., Wink D., & McIntosh M. Simulating Conditions of Slope Failure in the Lab with Sand and Water Using a Multimodal Monitoring Device for Data Collection. Poster session presented at: WiSys SPARK Symposium; 2023, August 8th, Tuesday; Oshkosh, WI.
- **Patino Luna H.** Constraining Landslides Simulation Models by Collecting Experimental Data of Water Movement Through Sand. Poster session presented at: 10th Annual King Chavez Scholars Research Poster Symposium; 2023, May 4th, Thursday; Whitewater, WI.
- Stephens B., **Patino Luna H**., Wink D., McIntosh M. Davis K., Larson C., & Lambert J. Physical Modeling with Sand and Water to Simulate Landslide Conditions to Predict Slope Failure Using Raspberry Pi. Poster session presented at: Fall 2023 Research Day Symposium; 2023, March 23rd, Thursday; Whitewater, WI.

INTERESTS & ACTIVITIES

In my free time I weightlift regularly and play soccer and pickleball occasionally. Enjoying rock, metal, and blues music has led me to trying to learn how to play electric and acoustic guitar. I have been working to improve my Spanish by reading in Spanish, including 1984 by George Orwell. I also enjoy fixing whatever breaks in my car and doing maintenance. Recent projects have been changing components of the breaks and suspension systems, oil changes, and welding exhaust and body components because of rust.